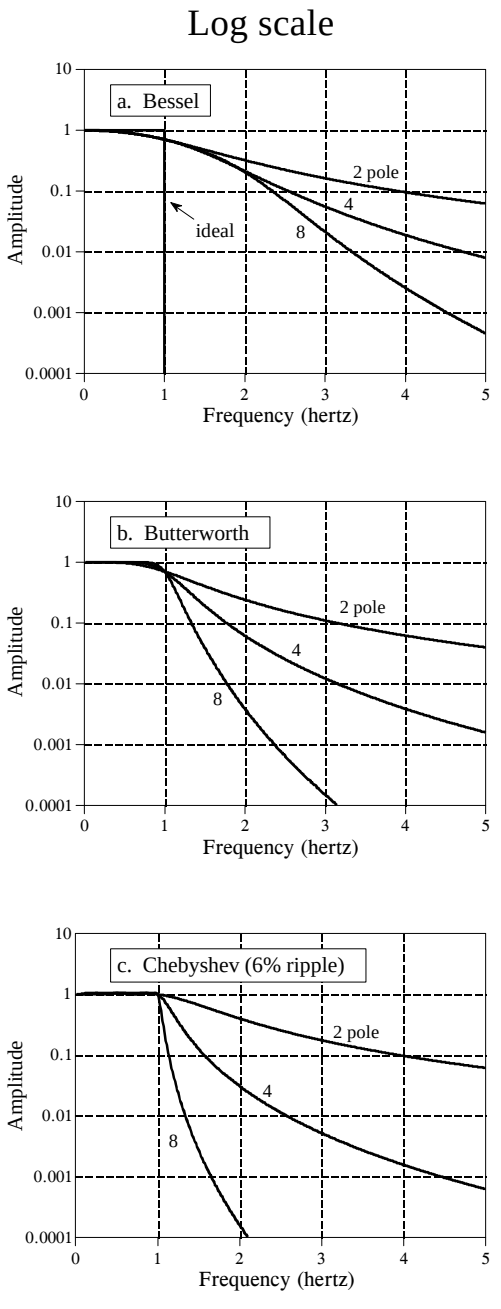


FIGURE 3-11
Frequency response of the three filters on a *logarithmic* scale. The Chebyshev filter has the sharpest roll-off.

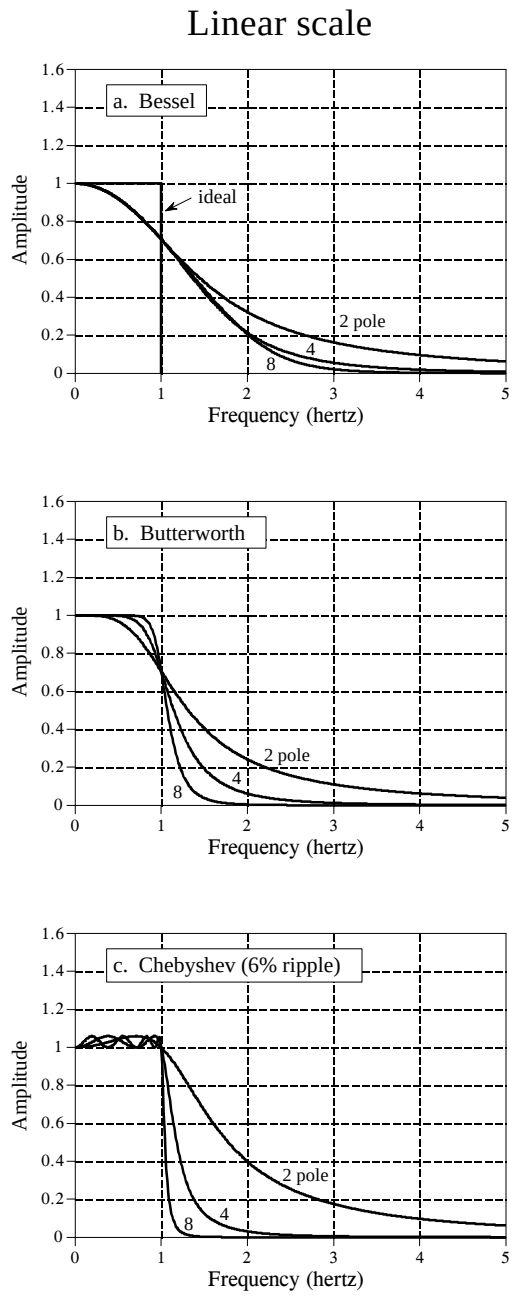


FIGURE 3-12
Frequency response of the three filters on a *linear* scale. The Butterworth filter provides the flattest passband.

in Fig. 3-12. **Passband ripple** can now be seen in the Chebyshev filter (wavy variations in the amplitude of the passed frequencies). In fact, the Chebyshev filter obtains its excellent roll-off by *allowing* this passband ripple. When more passband ripple is allowed in a filter, a faster roll-off